Lab 3 – Chromaticity: Building the CIE Diagram from Three Cones

CV8508 – Perceptual Cognition for Computer Vision

Lecture 5 showed the CIE diagram as a picture of all colours we can see. You will build a similar one from three bell-shaped cone sensitivities. Use of coding agents is allowed; write the explanations yourself.

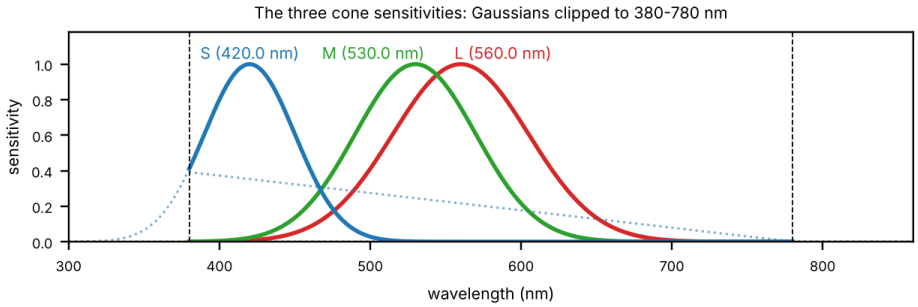
Setup

```
mkdir lab3 && cd lab3
wget https://utkarshmall13.github.io/cv8508-labs/lab3/setup.sh
sh setup.sh && source .venv/bin/activate     # lab3_tools.py, .venv
```

Read `lab3_tools.py`: `cone_sensitivities()` returns the L, M and S sensitivities (Gaussians in wavelength with peaks at 560, 530 and 420 nm, zero outside 380–780 nm, as in the figure below) and `lms_to_srgb()` turns an LMS triple into an approximate screen colour. A "light of wavelength λ" means unit power; mixing adds LMS vectors; E is the flat (equal-energy) spectrum, our white.

Tasks (each: output, then explain in 1–2 sentences)

Q1 Cones and locus. Plot the three sensitivities. Then compute LMS(λ) in small, discrete steps of wavelength (for example 5 nm) and draw the points as a 3D curve with axes L, M, S. A 3D plot is easy to misread from one angle, so show two different views, and label a few wavelengths. *Explain:* which cones respond at the two ends of the curve, and what does that say about the colours there?



Q2 Convex hull. Every light is a non-negative mix of monochromatic lights. Draw the convex hull of the locus *and the origin*, each face coloured by its LMS colour, from two different views, with a few wavelengths labelled. *Explain:* how did you get the convex hull of a set of points, why is the set of producible lights convex, and what does distance from the origin mean?

Q3 Slice. Cut the hull with $L + M + S = 1$ and plot it in $l = L/(L + M + S)$, $m = M/(L + M + S)$: labelled pure wavelengths, the line of purples, E, and a coloured interior (`lms_to_srgb` works on the vector $(l, m, 1 - l - m)$). *Explain:* what does the slice keep and discard, and how does it differ from the Lecture 5 diagram?

Q4 Primaries around white. Compute the coordinates (l, m) of E from the sensitivities, and also read them off the plot you drew in Q3; do the two agree? Mark red R at $(0.68, 0.25)$ and green G at $(0.40, 0.57)$ on the diagram (both inside the horseshoe), and choose a blue B such that the centroid of the triangle RGB is E. Check that B is inside the diagram, solve $r\,R + g\,G + b\,B = E$ (treat each point as the LMS vector $(l, m, 1 - l - m)$) and draw the triangle with E. *Explain:* how much of each primary makes E, and why does putting the centroid at E fix B?

Q5 What can the triangle show? Approximate the area of the whole diagram (horseshoe plus line of purples) that lies inside your triangle RGB and the area that lies outside it. Hint: put a dense, uniform grid of points over the (l, m) plane and count with a program how many fall inside the horseshoe and how many inside the triangle (or integrate numerically along l or m). What fraction of the colours in the diagram cannot be made with these three primaries? *Explain:* which colours are missing, and why is an area fraction in this diagram only a rough measure?